Adaptive Routing for Vector and Graph-Based Retrieval-Augmented Generation

by

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Declaration of Authorship

This report, Adaptive Routing for Vector and Graph-Based Retrieval-Augmented Generation, is submitted in partial fulfillment of the requirements of Master of Science in Artificial Intelligence at Munster Technological University Cork. I, William Barrett, declare that this thesis titled, Adaptive Routing for Vector and Graph-Based Retrieval-Augmented Generationand the work represents substantially the result of my own work except where explicitly indicated in the text. This report may be freely copied and distributed provided the source is explicitly acknowledged. I confirm that:

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- Where any part of this thesis has previously been submitted for a degree or any other qualification at Munster Technological University Cork or any other institution, this has been clearly stated.
- Where I have consulted the published work of others, this is always clearly attributed.
- Where I have quoted from the work of others, the source is always given. With the exception of such quotations, this project report is entirely my own work.
- I have acknowledged all main sources of help.
- Where the thesis is based on work done by myself jointly with others, I have made clear exactly what was done by others and what I have contributed myself.

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Abstract

Faculty of Engineering and Science
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Master of Science

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Retrieval-Augmented Generation is how we dynamically integrate external knowledge in to our agentic systems. LLMs work well on questioning corpuses of data but hallucinate when met with information they are not trained on. RAG is a proven solution to this but only performs well in specific chunk based lookups. RAG fails to at looking at thematic data across an entire corpus. In 2024 graphrag was proposed as a solution to this. GraphRag prebuilds a graph model identifying entities, communities and relationships across an entire corpus. Modern research focuses on the integration of vector or graph rag in isolation. A current gap is that there is little to no research looking at how to integrate both simultaneously. Current research takes an approach of splitting retrieval 50/50 or binary classification of vector or graph-based. The purpose of this research is to build and emperically evaluate routing strategies to address this gap. these straties will analyse incoming queries and determine an adaptive routing strategy that determines the best means of blending the results of both vector and graph based RAG. The contribution of this research will include the development and analysis of multiple classification and blending strategies, combined with an investigation into if adaptive routing out performs indivual retrievaal strategies.

Acknowledgements

The acknowledgements and the people to thank go here, don't forget to include your project supervisor...

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Abbreviations

LAH **L**ist **A**bbreviations **H**ere

For/Dedicated to/To my...

Chapter 1

Research Context

With the widespread adoption of large language models (LLMs), whose knowledge is limited to the parametric memory encoded during training, Retrieval-Augmented Generation (RAG) [1] has emerged as the predominant approach in enriching model outputs with non-parametric memory from external sources. Vector RAG encodes documents as chunks in the form of embeddings and retrieves the top k documents via semantic similarity given a query and adds them as context to the LLM for generation.

While vector RAG achieved state of the art performance in query focused summarization (QFS), Edge et al. [2] highlighted that it falls short when it comes to global or sense making queries like 'what are the main themes in this dataset?'. They proposed GraphRag as a solution to this problem. GraphRag adopts a graph-based approach to question answering. An LLM is first used to extract entities and relationships from source documents and construct a knowledge graph. Communities are then identified within the graph and summarised, this enables the system to answer global queries. However, GraphRAG has its own limitations. Graph construction introduces an expensive upfront cost, and for simple query-focused summarisation, vector RAG is far more efficient.

This highlights the need for adopting hybrid approaches to extending an LLM's knowledge. Xu et al. [3] tested this by deploying a hybrid model to production for six months. They took the approach of retrieving both vector and graph-based data and merging them in a single context. This hybrid approach was shown to reduce the median per-issue resolution time by 28.6%.

Recent works surrounding the area of hybrid approaches include Adaptive RAG proposed by Jeong et al. [4]. Adaptive RAG trains a classifier to predict the complexity of an

incoming query and classify whether to use no retrieval, single-step retrieval, or multi-step iterative retrieval. However, this works purly against vector-based RAG. Guo et al. [5] uses reinforcement learning to route between text and grapj retrieval, but the routing is limited to three options; vector, graph or both combined using recipricol rank fusion (RRF) [6].

This research investigates approaches to adpative routing bewteen vector and graph-based retrieval. This includes comparing rule-based, classifer-based and LLM-based methods to determine a weighting system that controls how much prompt space is allocation to each retrival types for a given query. This allows the system to optimize its retrieved context dynamically based on a query rather sticking to fixed retrieval strategies.

Chapter 2

Literature Review

The key question to answer in this chapter is: "What has been done/is being done".

You must present a summary of background research - what has been done in this area; any related work. Provide reader with information to understand the rest of the report. This chapter comprises around 3000 words and should put your project into context within the field. You must identify the main sources of information you should be aware of within your chosen area and use this activity to strengthen your knowledge in this area. So here you should develop a knowledge of not only your specific topic but also how it is changing.

Include relevant references. For the (rare) projects with no implementation component, e.g., using neural networks to explore or extend the properties of some mathematical model, or to make predictions of some kind, one might break this chapter into two and write a chapter on neural networks and another on the area of application.

To conduct this activity you should aim to read most of a relevant paper every other day for the remainder of the project, make a brief summary of the main findings and how they relate to your work (less than 30 words). You should continue this activity after your project as CPD.

You should begin your journey of discovery in reverse order to the listing above (which is given in order of academic importance/significance). So when you are researching your topic first look up some TedX talks or youtube tutorials, then research what companies are doing in the area, then get a handful of very good texts on the core topics of your area (anything older than 5 years usually is not helpful here) and finally start reading conference or journal papers (again newer is better here). In particular during this section you may need to use tables to list resources. These are also automatically formatted in latex thus allowing you to concentrate on content. for example table 2.1.

Parameter	PET
Youngs Modulus	2800-3100MPa
Tensile Strength	55-75MPa
Glass Temperature	75°C
Density	1400kg/m^3
Thermal Conductivity	0.15-0.24Wm^{-1}K^{-1}
Linear Expansion Coefficient	7×10^-5
Relative Dielectric Constant @ 1MHz	3
Dielectric Breakdown Strength	17kVmm^{-1}

Table 2.1: PET Physical Properties

Make sure you cite[7] as many **relevant** papers [8], luckily Latex will take care of the ordering of the citations [9] for you.

The aim here is that you find the trends in your topic (3), and more in general in the area in which your topic resides (2) your project falls under and from these trends you develop your initial project question further and begin to get insights into how others have solved/approached similar problems. Think of this section as colouring in your initial idea. Before you approach this section you should read at least 4/5 good review papers. Also remember if you find that you are detailing topics that you have not introduced earlier in this section you need to add a reference for background reading.

Chapter 3

Research Objectives

1. Implement a hybrid RAG system that can perform both vector and graph-based retrieval.
2. Develop a query classification framework to weigh incoming queries on the blend of vector to graph.
3. Analyse when and why adaptive blending outperforms fixed strategies.

Chapter 4

Implementation

This chapter should include the most relevant/important code listing and discuss relevant/interesting aspects; explain features you wish to highlight.

4.1 Important Features

For each aspect you could provide the following info:

1. Description of the difficulty: Brief description of the problem you found.
2. How did it affect the original project design?: Indicate how this difficulty affected:
   (a) the architecture of your solution
   (b) if it represented a risk to your overall project (i.e. was it a FR)
   (c) if it affected your methodology to develop your project (i.e. did developing this aspect change the design)
   (d) if it changed your implementation schedule (i.e. did it delay other items)
   (e) if it changed the evaluation plan (i.e. was more/less testing now required?)
   (f) If there is complex code/configuration details discuss it here also.

From January to September you will be developing your project. Along the way you will have encountered difficulties which might have modified your original plan so that you can come up with an actual developed project. You can discuss some of these that led to significant changes (if they exist). It is important here is that there is a clear rationale for design and implementation decisions made.

Chapter 5

Testing and Evaluation

The goal of this chapter is an objective evaluation of the final system. The evaluation must be quantitative and not qualitative. Indicate how the system was verified/tested. You may perform qualitative evaluation but this should not form the basis of the main conclusions you derive from the evaluation. This evaluation, where possible, should be comparative, i.e. you should evaluate your system against a commercially available system and/or system detailed in a research publication. If no implementation was done, you could describe the types of experiments/simulations that were carried out. Explain why these and not others were performed and discuss the most important parameters in the tests/simulations and describe their roles.

You should demonstrate operational testing of the project using real or contrived data sets to evaluate aspects of the project not encompassed in the software testing (e.g. quantify how well does your project achieved the overall goal).

- For software based projects this will include, but should not be limited to, evaluation of non-functional requirements.
- For infrastructural projects this testing should include system/network KPI analysis.
- For analysis based projects (ML, malware or other) this may include model evaluation or YARA rule validation, for example.
- For management projects, where software testing or infrastructure testing may not be in scope, the test process for the system is expected to be more rigorous and well described than a project incorporating significant development work.

Summarize the output data, and the statistical or other techniques to deduce your results. Summarize your results, including tables or graphs as appropriate with a brief

description of each. Here, where possible, compare your results with other products/systems. Identify any possible threats to the validity of your results, and discuss each briefly here (you will discuss in more detail in the next chapter).

Chapter 6

Discussion and Conclusions

Write a critical evaluation of your work. You should state your goals and research questions again and outline in short how they were or were not met or answered. What have you achieved? What more could have been accomplished? Mention any problems that affected the progress of your work. Is there any scope for extension/improvement/scalability? What are the new challenges to be faced in the future? The expectation here is that you should discuss the results presented in the previous evaluation section of the project in their totality (i.e. as a whole) from which you will then draw clear conclusions both on the quantitative and qualitative aspects of the overall project. The conclusion section of this chapter should conclude the project.

6.1 Discussion

Discuss how well your solution solves the problem, based on your results from the evaluation chapter. Discuss how well you addressed the project, and what you might do differently if you were to do it again. Make sure to identify how you handled any problems that arose during the project.

6.2 Conclusion

Enumerate the main conclusions you have got in terms of background, problem description and the solution approach you have come up with. Detail your primary and any secondary conclusions from your project.

6.3 Future Work

Discuss any proposals for completion of the project, or for enhancements, or for re-design of your solution or software. Enumerate all the things you would have wanted to do should you have more time to work on this project.

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Appendix A

Code Snippets

Put appendix material in this section e.g. code snippets

If there is too much code to include in the Appendix, submit the rest on a USB disk or CD. Some code should be easily available for examiners to assess.

USE THE APPENDICES

Appendix B

Wireframe Models